Task 02 - Packet Sniffing, Firewall Configuration and Vulnerability Scanning Report

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> **Date:** February 19, 2026
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> **Position:** VAPT Intern

1. Objective

The goal of this lab was to gain hands-on experience with:

- Network packet capture and basic protocol analysis using **Scapy**
- Basic stateful firewall configuration using `iptables`
- Vulnerability scanning and result interpretation using **OpenVAS / Greenbone Vulerability Manager**

2. Tools Used

Task	Tool / Software	Version / Note
Packet Sniffing	Scapy	Python library (New version)
Visualization	`metplotlib`	Python plotting library
Firewall	`iptables`	Linux kernel netfilter
Vulerability Scanning	Greenbone Community Edition	OpenVAS / GVM (little bit old version - from official docker container)
Operating System	Arch Linux	localhost / 127.0.0.1 scanned

3. Task 1 - Packet Sniffing with Scapy

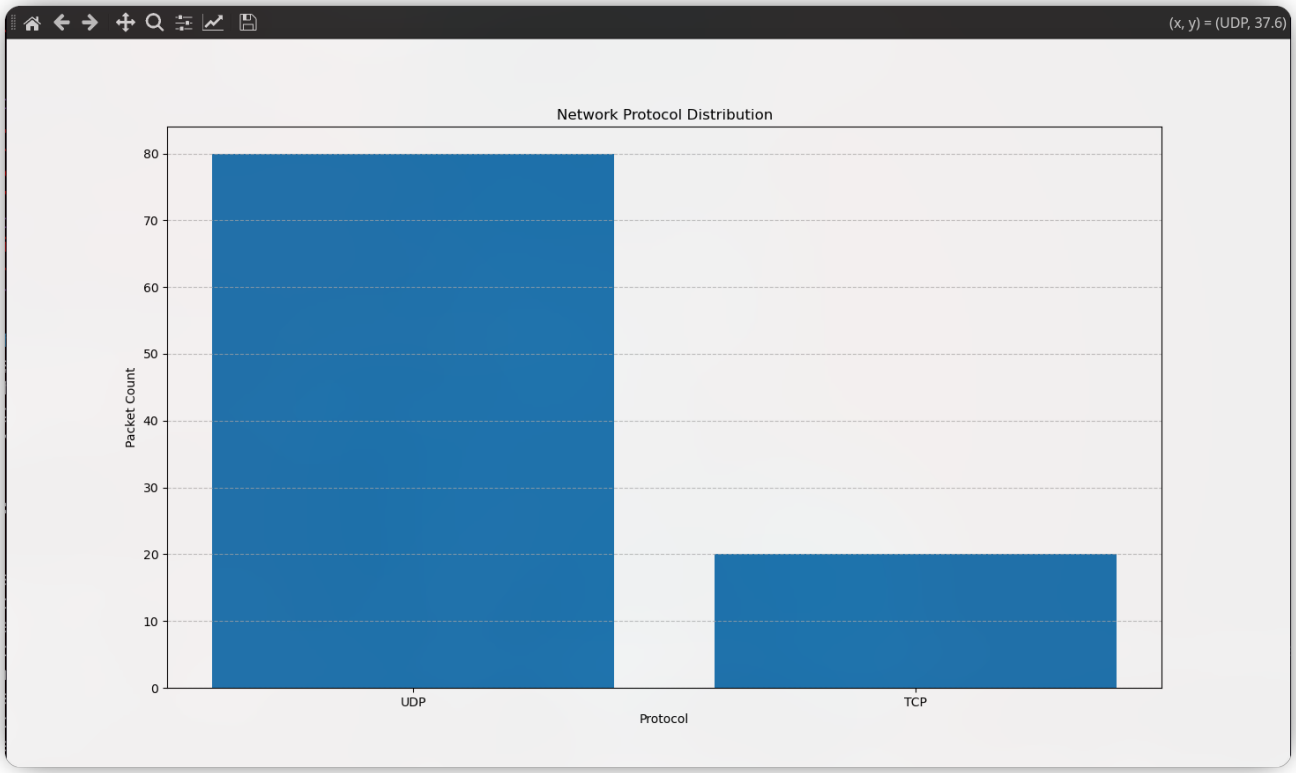
3.1 Script Used

- Script used can be access here: `scapy_scanner.py`

3.2 Results

Captured on **loopback interface (`lo`)**, mostly local traffic

Visual Bar chart generated by the `scapy_scanner.py`:



Results:

```
Protocol Distribution:
+ UDP: 80
- TCP: 20
```

Most traffic on loopback in a quiet system tends to be UDP (DNS, mDNS, systemd, CUPS, Avahi, etc.) rather than TCP.

4. Task 2 - Basic `iptables` Firewall

4.1 Rules implemented

```
# 1. Flush existing rules
sudo iptables -F
sudo iptables -X

# 2. Set default policies
sudo iptables -P INPUT DROP
sudo iptables -P FORWARD DROP
sudo iptables -P OUTPUT ACCEPT

# 3. Allow loopback (*imp)
sudo iptables -A INPUT -i lo -j ACCEPT
sudo iptables -A OUTPUT -o lo -j ACCEPT
```

```
# 4. Allow estabilished & related connections
sudo iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT

# 5. Allow SSH (port 22)
sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT

# 6. Allow HTTP (port 80) – optional in this lab
sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT

# 7. Optional: allow ping (ICMP echo-request)
sudo iptables -A INPUT -p icmp --icmp-type echo-request -j ACCEPT
```

4.2 Verification

```
# Check rules
sudo iptables -L -v -n --line-numbers

# Test: SSH should work
# Test: HTTP should work
# Test: HTTPS should work
# Test: random high port should be blocked
```

Result: *Firewall correctly blocked all unsolicated inbound traffic expert SSH (and optionally HTTP/HTTPS)*

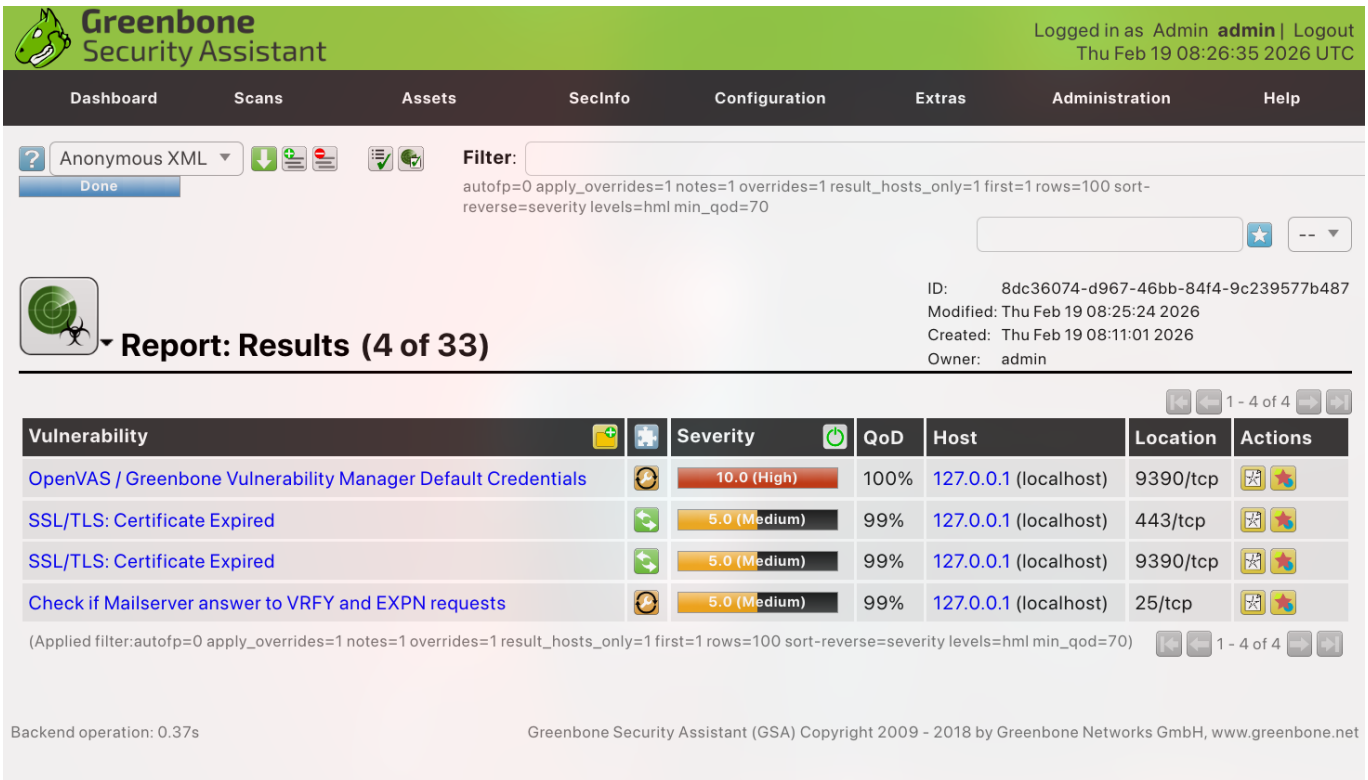
5. Task 3 - OpenVAS / Greenbone Vulnerability Scan

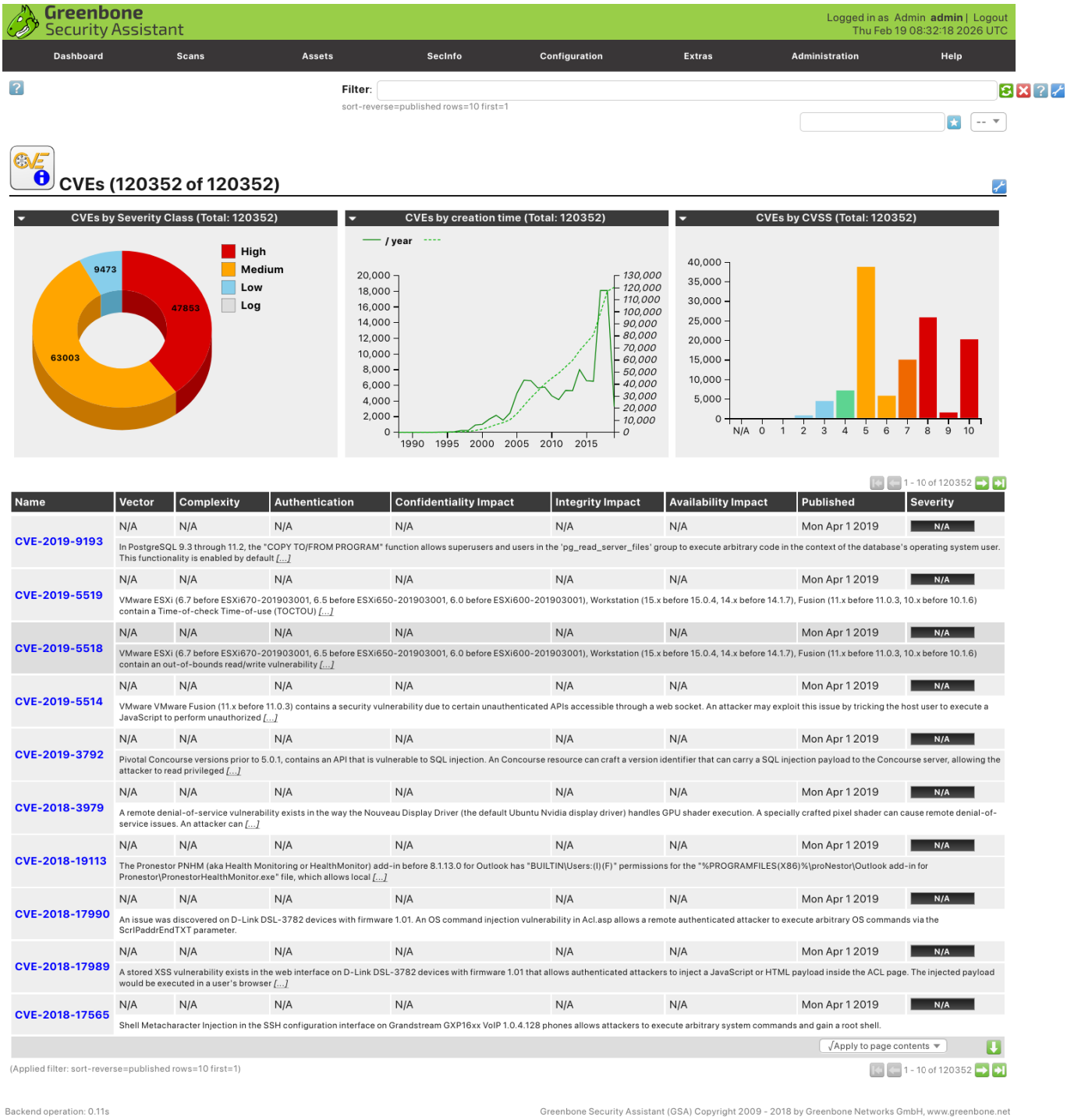
5.1 Target

- **Scanned host:** 127.0.0.1 (`localhost`)
- **Tool:** OpenVAS running in a docker container on localhost

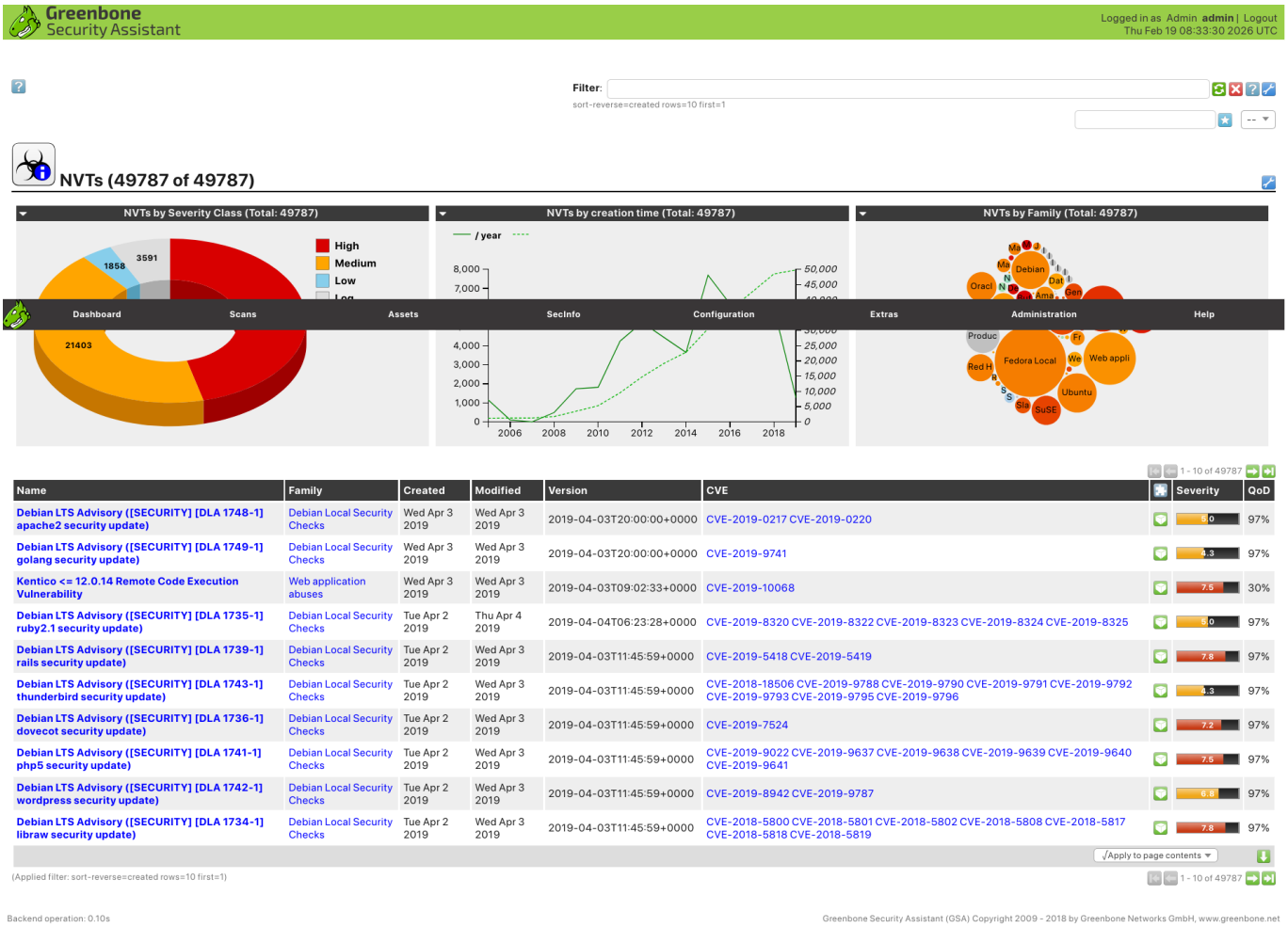
5.2 Key Findings

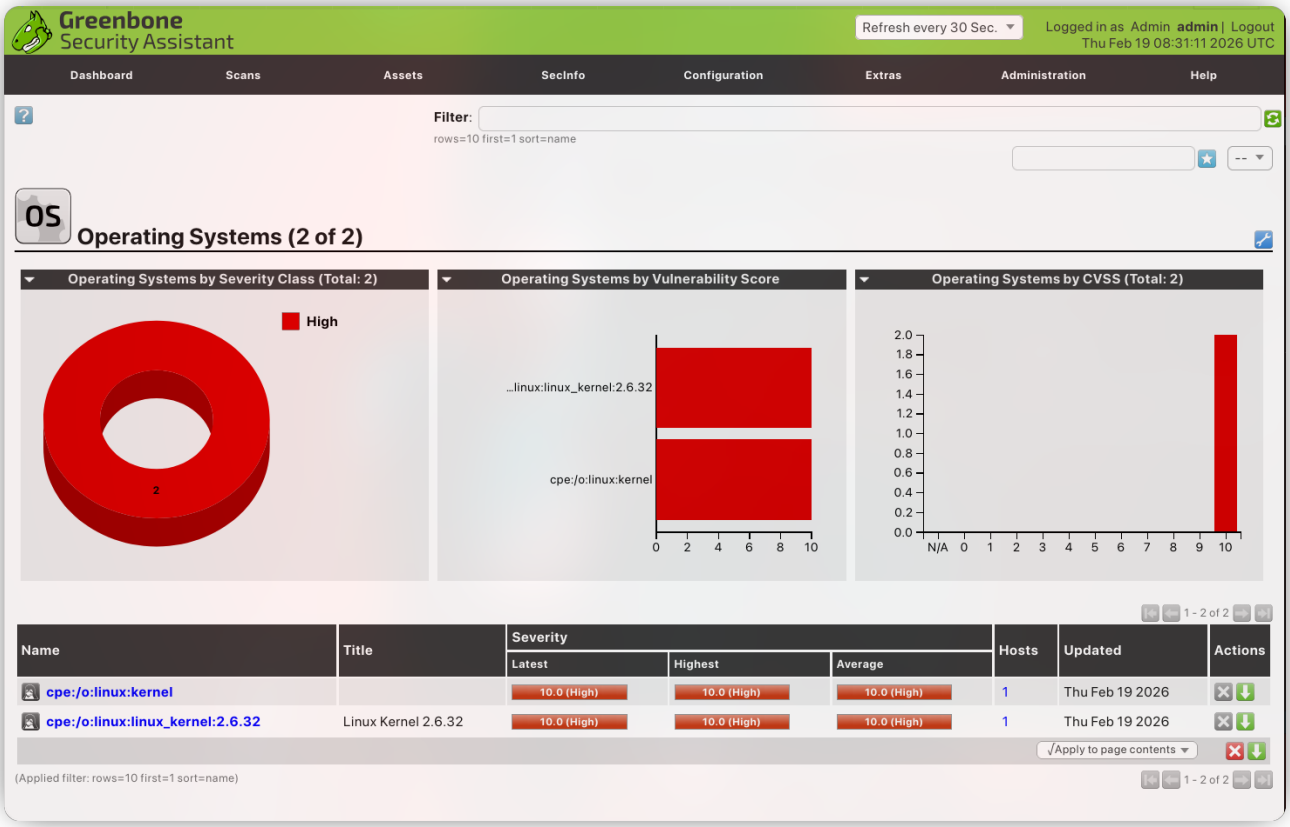
Greenbone Security Assistant
Refresh every 30 Sec.
Logged in as Admin admin | Logout
Thu Feb 19 08:29:15 2026 UTC
Dashboard Scans Assets SecInfo Configuration Extras Administration Help
Filter:
min_qod=70 apply_overrides=1 rows=10 first=1 sort=name
Tasks (1 of 1)
Tasks by Severity Class (Total: 1)
High
1
Tasks with most High results per host
Immediate scan of IP 127...
0.0 0.2 0.4 0.6 0.8 1.0
Tasks by status (Total: 1)
Done
1
1 - 1 of 1
Name Status Reports Total Last Severity Trend Actions
Immediate scan of IP 127.0.0.1 Done 1 (1) Feb 19 2026 10.0 (High)
Apply to page contents
(Applied filter: min_qod=70 apply_overrides=1 rows=10 first=1 sort=name)
1 - 1 of 1
Backend operation: 0.01s
Greenbone Security Assistant (GSA) Copyright 2009 - 2018 by Greenbone Networks GmbH, www.greenbone.net

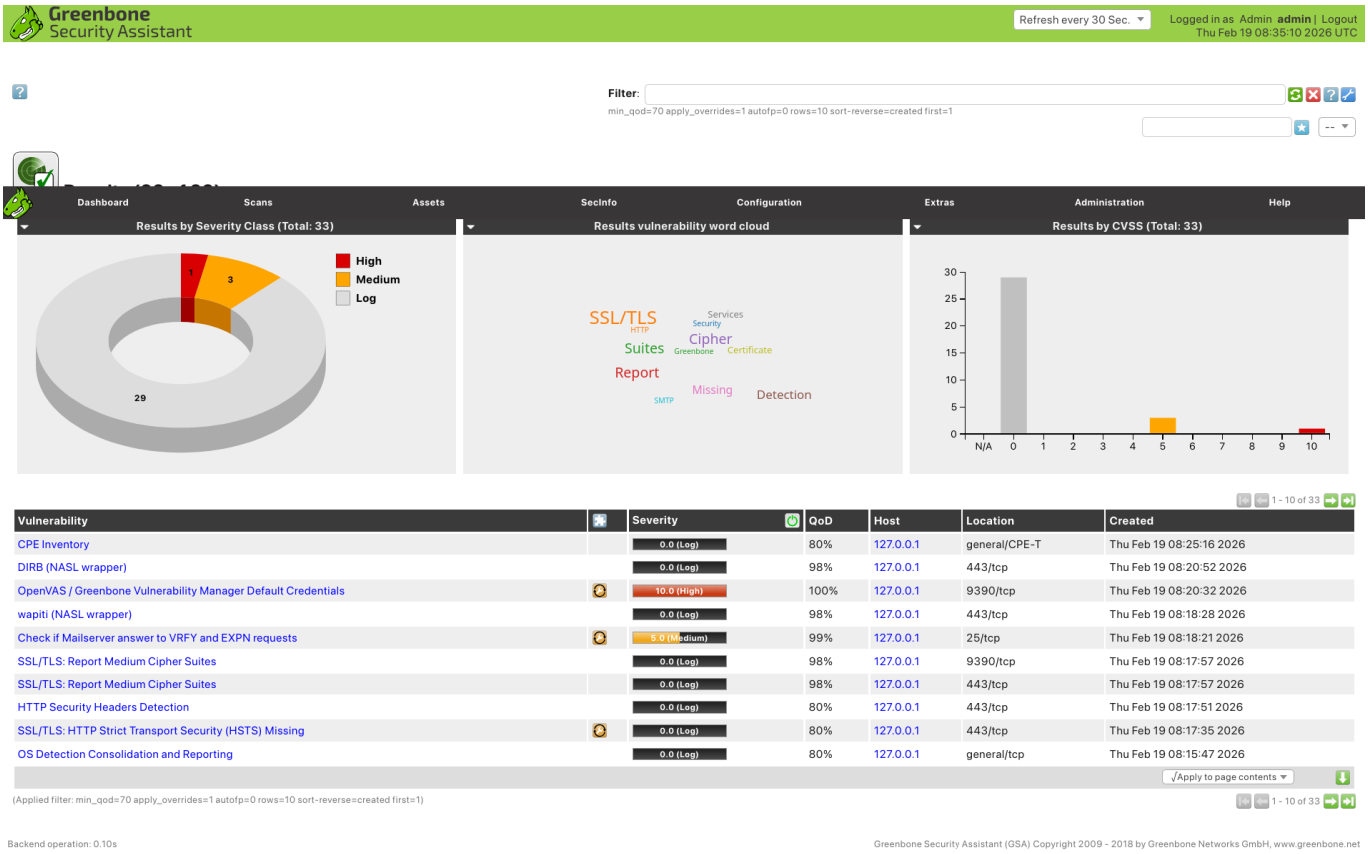

Greenbone Security Assistant
Logged in as Admin admin | Logout
Thu Feb 19 08:26:35 2026 UTC
Dashboard Scans Assets SecInfo Configuration Extras Administration Help
Anonymous XML
Done
Filter:
autofp=0 apply_overrides=1 notes=1 overrides=1 result_hosts_only=1 first=1 rows=100 sort-reverse=severity levels=hml min_qod=70
Report: Results (4 of 33)
ID: 8dc36074-d967-46bb-84f4-9c239577b487
Modified: Thu Feb 19 08:25:24 2026
Created: Thu Feb 19 08:11:01 2026
Owner: admin
1 - 4 of 4
Vulnerability Severity QoD Host Location Actions
OpenVAS / Greenbone Vulnerability Manager Default Credentials 10.0 (High) 100% 127.0.0.1 (localhost) 9390/tcp
SSL/TLS: Certificate Expired 5.0 (Medium) 99% 127.0.0.1 (localhost) 443/tcp
SSL/TLS: Certificate Expired 5.0 (Medium) 99% 127.0.0.1 (localhost) 9390/tcp
Check if Mailserver answer to VRFY and EXPN requests 5.0 (Medium) 99% 127.0.0.1 (localhost) 25/tcp
(Applied filter:autofp=0 apply_overrides=1 notes=1 overrides=1 result_hosts_only=1 first=1 rows=100 sort-reverse=severity levels=hml min_qod=70)
1 - 4 of 4
Backend operation: 0.37s
Greenbone Security Assistant (GSA) Copyright 2009 - 2018 by Greenbone Networks GmbH, www.greenbone.net

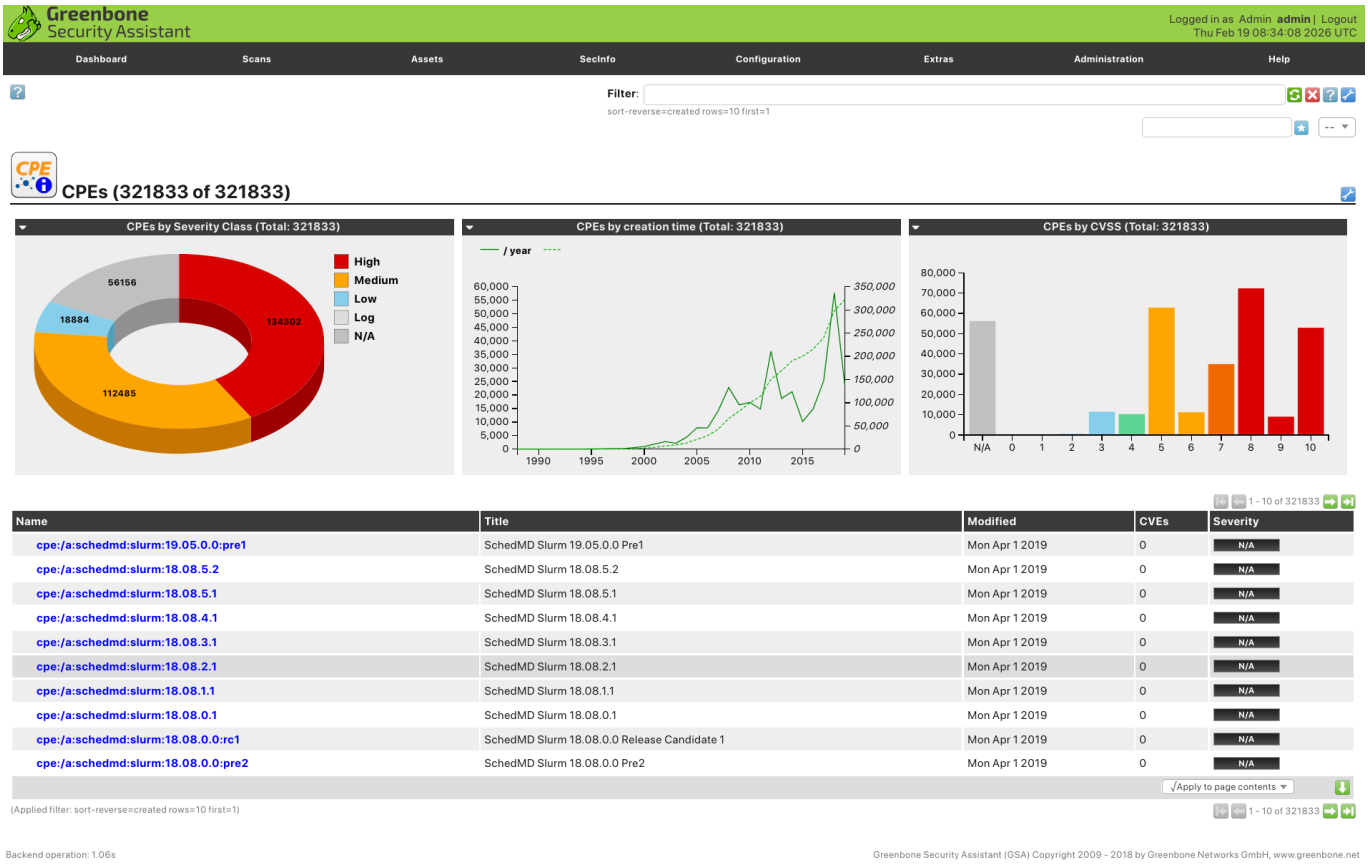


Name	Vector	Complexity	Authentication	Confidentiality Impact	Integrity Impact	Availability Impact	Published	Severity
CVE-2019-9193	N/A	N/A	N/A	N/A	N/A	N/A	Mon Apr 1 2019	N/A
	In PostgreSQL 9.3 through 11.2, the "COPY TO/FROM PROGRAM" function allows superusers and users in the 'pg_read_server_files' group to execute arbitrary code in the context of the database's operating system user. This functionality is enabled by default [...]							
CVE-2019-5519	N/A	N/A	N/A	N/A	N/A	N/A	Mon Apr 1 2019	N/A
	VMware ESXi (6.7 before ESXi670-201903001, 6.5 before ESXi650-201903001, 6.0 before ESXi600-201903001), Workstation (15.x before 15.0.4, 14.x before 14.1.7), Fusion (11.x before 11.0.3, 10.x before 10.1.6) contain a Time-of-check Time-of-use (TOCTOU) [...]							
CVE-2019-5518	N/A	N/A	N/A	N/A	N/A	N/A	Mon Apr 1 2019	N/A
	VMware ESXi (6.7 before ESXi670-201903001, 6.5 before ESXi650-201903001, 6.0 before ESXi600-201903001), Workstation (15.x before 15.0.4, 14.x before 14.1.7), Fusion (11.x before 11.0.3, 10.x before 10.1.6) contain an out-of-bounds read/write vulnerability [...]							
CVE-2019-5514	N/A	N/A	N/A	N/A	N/A	N/A	Mon Apr 1 2019	N/A
	VMware VMware Fusion (11.x before 11.0.3) contains a security vulnerability due to certain unauthenticated APIs accessible through a web socket. An attacker may exploit this issue by tricking the host user to execute a JavaScript to perform unauthorized [...]							
CVE-2019-3792	N/A	N/A	N/A	N/A	N/A	N/A	Mon Apr 1 2019	N/A
	Pivotal Concourse versions prior to 5.0.1, contains an API that is vulnerable to SQL injection. An Concourse resource can craft a version identifier that can carry a SQL injection payload to the Concourse server, allowing the attacker to read privileged [...]							
CVE-2018-3979	N/A	N/A	N/A	N/A	N/A	N/A	Mon Apr 1 2019	N/A
	A remote denial-of-service vulnerability exists in the way the Nouveau Display Driver (the default Ubuntu Nvidia display driver) handles GPU shader execution. A specially crafted pixel shader can cause remote denial-of-service issues. An attacker can [...]							
CVE-2018-19113	N/A	N/A	N/A	N/A	N/A	N/A	Mon Apr 1 2019	N/A
	The Pronestor PNHM (aka Health Monitoring or HealthMonitor) add-in before 8.1.13.0 for Outlook has "BUILTIN\Users:(I)(F)" permissions for the "%PROGRAMFILES(X86)%\proNestor\Outlook add-in for Pronestor\PronestorHealthMonitor.exe" file, which allows local [...]							
CVE-2018-17990	N/A	N/A	N/A	N/A	N/A	N/A	Mon Apr 1 2019	N/A
	An issue was discovered on D-Link DSL-3782 devices with firmware 1.01. An OS command injection vulnerability in Acl.asp allows a remote authenticated attacker to execute arbitrary OS commands via the ScrIPaddrEndTXT parameter.							
CVE-2018-17989	N/A	N/A	N/A	N/A	N/A	N/A	Mon Apr 1 2019	N/A
	A stored XSS vulnerability exists in the web interface on D-Link DSL-3782 devices with firmware 1.01 that allows authenticated attackers to inject a JavaScript or HTML payload inside the ACL page. The injected payload would be executed in a user's browser [...]							
CVE-2018-17565	N/A	N/A	N/A	N/A	N/A	N/A	Mon Apr 1 2019	N/A
	Shell Metacharacter Injection in the SSH configuration interface on Grandstream GXP16xx VoIP 1.0.4.128 phones allows attackers to execute arbitrary system commands and gain a root shell.							


Greenbone Security Assistant
Logged in as Admin admin | Logout
Thu Feb 19 08:33:30 2026 UTC
Filter:
sort-reverse=created rows=10 first=1
NVTs (49787 of 49787)
NVTs by Severity Class (Total: 49787)
High
Medium
Low
3591
1858
21403
NVTs by creation time (Total: 49787)
/ year
NVTs by Family (Total: 49787)
Dashboard
Scans
Assets
SecInfo
Configuration
Extras
Administration
Help
1 - 10 of 49787
Name
Family
Created
Modified
Version
CVE
Severity
QoD
Debian LTS Advisory ([SECURITY] [DLA 1748-1] apache2 security update)
Debian Local Security Checks
Wed Apr 3 2019
Wed Apr 3 2019
2019-04-03T20:00:00+0000
CVE-2019-0217 CVE-2019-0220
5.0
97%
Debian LTS Advisory ([SECURITY] [DLA 1749-1] golang security update)
Debian Local Security Checks
Wed Apr 3 2019
Wed Apr 3 2019
2019-04-03T20:00:00+0000
CVE-2019-9741
4.3
97%
Kentico <= 12.0.14 Remote Code Execution Vulnerability
Web application abuses
Wed Apr 3 2019
Wed Apr 3 2019
2019-04-03T09:02:33+0000
CVE-2019-10068
7.5
30%
Debian LTS Advisory ([SECURITY] [DLA 1735-1] ruby2.1 security update)
Debian Local Security Checks
Tue Apr 2 2019
Thu Apr 4 2019
2019-04-04T06:23:28+0000
CVE-2019-8320 CVE-2019-8322 CVE-2019-8323 CVE-2019-8324 CVE-2019-8325
5.0
97%
Debian LTS Advisory ([SECURITY] [DLA 1739-1] rails security update)
Debian Local Security Checks
Tue Apr 2 2019
Wed Apr 3 2019
2019-04-03T11:45:59+0000
CVE-2019-5418 CVE-2019-5419
7.8
97%
Debian LTS Advisory ([SECURITY] [DLA 1743-1] thunderbird security update)
Debian Local Security Checks
Tue Apr 2 2019
Wed Apr 3 2019
2019-04-03T11:45:59+0000
CVE-2018-18506 CVE-2019-9788 CVE-2019-9790 CVE-2019-9791 CVE-2019-9792 CVE-2019-9793 CVE-2019-9795 CVE-2019-9796
4.3
97%
Debian LTS Advisory ([SECURITY] [DLA 1736-1] dovecot security update)
Debian Local Security Checks
Tue Apr 2 2019
Wed Apr 3 2019
2019-04-03T11:45:59+0000
CVE-2019-7524
7.2
97%
Debian LTS Advisory ([SECURITY] [DLA 1741-1] php5 security update)
Debian Local Security Checks
Tue Apr 2 2019
Wed Apr 3 2019
2019-04-03T11:45:59+0000
CVE-2019-9022 CVE-2019-9637 CVE-2019-9638 CVE-2019-9639 CVE-2019-9640 CVE-2019-9641
7.5
97%
Debian LTS Advisory ([SECURITY] [DLA 1742-1] wordpress security update)
Debian Local Security Checks
Tue Apr 2 2019
Wed Apr 3 2019
2019-04-03T11:45:59+0000
CVE-2019-8942 CVE-2019-9787
6.8
97%
Debian LTS Advisory ([SECURITY] [DLA 1734-1] libraw security update)
Debian Local Security Checks
Tue Apr 2 2019
Wed Apr 3 2019
2019-04-03T11:45:59+0000
CVE-2018-5800 CVE-2018-5801 CVE-2018-5802 CVE-2018-5808 CVE-2018-5817 CVE-2018-5818 CVE-2018-5819
7.8
97%
Apply to page contents
(Applied filter: sort-reverse=created rows=10 first=1)
Backend operation: 0.10s
Greenbone Security Assistant (GSA) Copyright 2009 - 2018 by Greenbone Networks GmbH, www.greenbone.net


Greenbone Security Assistant
Refresh every 30 Sec.
Logged in as Admin admin | Logout
Thu Feb 19 08:31:11 2026 UTC
Dashboard
Scans
Assets
SecInfo
Configuration
Extras
Administration
Help
Filter:
rows=10 first=1 sort=name
Operating Systems (2 of 2)
Operating Systems by Severity Class (Total: 2)
High
2
Operating Systems by Vulnerability Score
...linux:linux_kernel:2.6.32
cpe:/o:linux:kernel
Operating Systems by CVSS (Total: 2)
1 - 2 of 2
Name
Title
Severity
Latest
Highest
Average
Hosts
Updated
Actions
cpe:/o:linux:kernel
10.0 (High)
10.0 (High)
10.0 (High)
1
Thu Feb 19 2026
cpe:/o:linux:linux_kernel:2.6.32
Linux Kernel 2.6.32
10.0 (High)
10.0 (High)
10.0 (High)
1
Thu Feb 19 2026
Apply to page contents
(Applied filter: rows=10 first=1 sort=name)





High Severity issue:

1. **OpenVAS / Greenbone Vulnerability Manager Default Credentials**

- Severity: **10.0 (High)**
- Port: 9390/tcp

- Impact: complete system compromise possible

2. **SSL/TLS Certificate Expired**

- Severity: **5.0 (Medium)**
- PortL 443/tcp, 9390/tcp

3. **Check if Mailserver answers to VRFY and EXPN requests**

- Severity: **5.0 (Medium)**
- Port: 25/tcp

4. **Various Log and Medium SSL/TLS issues**

- Missing HSTS
- Medium strength ciphers still offered
- No strict TLS configuration

Summary statistics (from dashboard of OpenVAS)

- Total results: **33**
- High: **1**
- Medium: **3**
- Log/None: 29

6. Key Learning & Challenges

- **Scapy on loopback** gives very biased protocol distribution (mostly UDP). Real scans should use physical interfaces.
- **iptables** default DROP policy + loopback exception is critical – forgetting it breaks almost everything.
- Many modern **OpenVAS/GVM** installations come with **default credentials** – this is one of the most common and dangerous findings.
- Expired/self-signed certificates are very frequent in internal/lab environments but still lower business impact than authentication bypasses.
- Reading OpenVAS reports takes practice – important fields are: **QoD (Quality of Detection)**, **CVSS score**, **solution type**, **insight**.

7. Recommendations for Improvment

- Change default OpenVAS/GVM admin credentials immediately
- Renew or replace expired certificates (or suppress false-positive if self-signed is intentional)
- Disable VRFY/EXPN on mail servers if present
- Strengthen TLS configuration (disable weak ciphers, enforce HSTS, prefer TLS 1.3)
- Capture packets on real network interface for next protocol analysis exercise

8. Sources Consulted

- Scapy official documentation — https://scapy.readthedocs.io/
- iptables tutorial – netfilter.org / digitalocean.com/community
- Greenbone Community Documentation — https://greenbone.github.io/docs/
- OpenVAS setup guides (Kali, Ubuntu) – various blog posts 2023–2025

End of Report